

**FLOODSENSE: AN ANDROID-BASED EXPERT SYSTEM FOR FLOOD
SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN
BACOR CITY**

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FloodSense: An Android-Based Expert System for Flood Susceptibility Assessment and Pre-Event Preparedness in Bacoor City

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INTRODUCTION

Flooding is one of the most frequent and destructive natural hazards in the Philippines. The country's location within the typhoon belt exposes communities to rainfall events that cause inundation, displacement, damage to property, interruption of livelihood, and loss of life. National disaster statistics continue to document the widespread effects of natural hazards on Filipino families and communities (Philippine Statistics Authority [PSA], 2026). Bacoor City, Cavite, is among the localities that regularly experience flooding. Its low-lying areas and connection to the Imus and Bacoor river systems contribute to persistent inundation during periods of heavy rainfall. Flood-mitigation facilities in the Imus River Basin were established specifically to reduce peak floodwater volume in Cavite, including low portions of Bacoor and its tributaries (Japan International Cooperation Agency [JICA], 2021). These conditions establish the importance of accessible, understandable, and location-specific information that residents can consult before a flood event develops.

To assess the information problem, the researchers surveyed 50 residents across 12 barangays in Bacoor City, with respondents drawn largely from historically flood-affected communities. The findings revealed that flood warnings are commonly

disseminated through Short Message Service (SMS), social media, and barangay public-address systems, yet these channels do not consistently provide location-specific susceptibility information before flooding occurs. In the first preliminary survey, 10 of 50 respondents felt well-informed about flood-prone areas; the other 40 selected Somewhat Aware, Not Sure, or Not Aware at All. The report's mean ratings for satisfaction, timeliness, and clarity were 2.62, 2.70, and 2.72 on a five-point scale, within its Neutral/Sometimes interpretation band. Thirty-two of 50 respondents reported making a wrong decision because of insufficient flood information. A key informant interview with the Bacoar Disaster Risk Reduction and Management Office (BDRRMO) described historical flood records and Geographic Information System (GIS) materials maintained institutionally. A supplementary interview with the City Planning and Development Coordinator Office (CPDCO) identified relevant elevation, hazard, land-use, and barangay-boundary maps, while also identifying limits on access to machine-readable spatial files.

A separate preliminary survey analysis report dated May 2026 summarized 40 respondents. It recorded 31 of 40 reporting flooding almost every rainy season, 9 of 40 considering themselves well-informed about flood-prone areas, and 26 of 40 selecting a dissatisfied category for available flood information. The report does not establish its exact collection dates, respondent-selection procedure, or overlap with the first survey; the two groups are therefore not pooled and their results are not treated as citywide estimates or final system-evaluation findings.

FloodSense addresses this information gap through an Android application intended to connect expert-validated knowledge with the pre-event information needs of Bacoar City residents. A deterministic Rule-Based Expert System uses structured geographic and rainfall facts and conditional rules to produce an explainable susceptibility class for a user-confirmed scenario, subject to the provenance and validation of its underlying data. Rule-based expert systems apply rules from a knowledge base to known facts to reach a conclusion (Buchanan & Shortliffe, 1984).

The current prototype demonstrates Low, Moderate, High, and Very High outputs using provisional or demonstration inputs; it does not yet establish approved operational classifications for Bacoor City. A separate Decision Support System (DSS) presents published preparedness guidance using the assessment as context without changing its class. Decision support systems organize relevant information for human judgment (Power, 2002); a full resident-facing situational questionnaire has not been verified as complete.

The current FloodSense prototype presents scenario-based susceptibility assessment through an interactive map. The user selects or confirms a hypothetical rainfall intensity and duration, selects a supported location manually or through an optional one-time foreground location request, and explicitly requests an assessment. The map can display demonstration susceptibility zones and a neutral reference layer representing the 47 current barangay identities; the reference layer is technically validated for geometry but is not City-issued, is not City-verified, and contains no flood-susceptibility facts. The custom Web Administration System currently provides authenticated dashboard access, read-only review of map and rainfall-reference data, parameter-governance views, and controlled workflows for Decision Support System content, evacuation-center records, source records, and audit history. Ordinary administrators do not edit raw Expert System rules, priorities, conflict-resolution logic, or algorithm code. A fixed-schema CSV export-and-import workflow for permitted values is a confirmed requirement, but it is not yet implemented. FloodSense does not provide real-time monitoring, official forecasting, autonomous alerts, or automatic evacuation orders.

Statement of the Problem

This study aimed to develop an Android application that provides location-specific flood susceptibility assessment and pre-event preparedness support for Bacoor City through a Rule-Based Expert System, a Decision Support System, and a

dynamic geographic map. The research questions were grounded in the survey of 50 residents across 12 barangays, key informant interviews with the BDRRMO and CPDCO, and a review of related literature and official information. These sources were used to establish community needs, institutional knowledge sources, geographic requirements, rainfall parameters, preparedness content, and the functional boundaries of the proposed application and its Web Administration System.

In Bacoor City, flooding is a persistent and consequential reality. Survey results reveal that 92% of respondents have personally experienced flooding, with 58% describing the flooding as severe or waist-deep and higher. Despite this exposure, only 20% consider themselves well-informed about flood-prone areas. Mean satisfaction scores for information quality, timeliness, and clarity ranged from 2.62 to 2.72 on a 5-point scale, clustering at the lower end of the Neutral range and tending toward dissatisfaction. These information deficiencies have direct consequences: 64% of respondents reported making wrong decisions due to insufficient flood information, including taking flooded routes, failing to evacuate on time, and misjudging conditions. Open-ended responses further identified late or delayed information at 26% and insufficient or absent location-specific warnings at 20% as the leading concerns. How can an Android application using an Expert System provide residents with understandable, location-specific flood susceptibility assessments based on expert-validated knowledge, rainfall scenarios, and geographic information?

The problem also concerns the accessibility and operational maintenance of institutional knowledge. The BDRRMO maintains historical flood records, exposure information, preparedness protocols, and evacuation-center information, while the CPDCO maintains geographic and planning data. Current dissemination practices, including social media, email, and barangay coordination, do not fully address the public's need for on-demand, situation-specific guidance. A community-facing platform must therefore communicate approved information while ensuring that rules, map objects, preparedness content, rainfall parameters, data sources, and administrative

accounts can be maintained by authorized personnel. How can a Rule-Based Expert System, dynamic GIS-style map, Decision Support System, and Web Administration System organize this institutional knowledge into an explainable assessment and appropriate pre-event preparedness guidance for Bacoar City residents?

Taken together, these gaps reveal a disconnect between institutional flood knowledge and the decisions residents must make before flooding occurs. The overarching question of the study is: How can FloodSense integrate expert-provided and expert-validated knowledge, rule-based inference, scenario-based rainfall input, interactive geographic visualization, Check My Area, evacuation-center information, and structured decision support to strengthen flood susceptibility awareness and pre-event preparedness in Bacoar City, while remaining a complementary public-information tool that does not perform real-time flood prediction or supersede official warnings?

Objectives of the Study

Generally, the study aimed to develop an Android application that uses a Rule-Based Expert System to assess flood susceptibility and a Decision Support System to provide pre-event preparedness support for residents of Bacoar City through explainable expert knowledge, rainfall scenarios, and dynamic geographic visualization.

Specifically, it aimed to:

1. identify the prevailing challenges, pain points, and needs of Bacoar City residents and relevant local government offices concerning flood information access, timeliness, clarity, location-specific awareness, and pre-event decision-making through preliminary resident surveys and key-informant interviews with the Bacoar Disaster Risk Reduction and Management Office (BDRMO) and the City Planning and Development Coordinator Office (CPDCO);

2. analyze expert-provided and expert-validated flood knowledge, BDRMO historical flood records and preparedness protocols, CPDCO geographic information, PAGASA rainfall classifications, evacuation-center information, and user situational requirements to determine the facts, conditional rules, susceptibility levels, rainfall parameters, geographic entities, and DSS content required by the knowledge base and inference process;
 3. develop an Android application using Flutter and Dart, supported by Django, Django REST Framework, GeoDjango, and PostgreSQL/PostGIS, that integrates a deterministic Rule-Based Expert System, scenario-based rainfall input, an interactive GIS-style map, manual location selection, optional one-time foreground location, explainable susceptibility output, and separate Decision Support System guidance; provide a custom Web Administration System for authorized review and maintenance of permitted content and records; protect raw Expert System rules and algorithm logic from ordinary administrator editing; and establish a fixed-schema CSV exchange requirement for permitted values without representing that requirement as operational until its routes, validation, interface, integration, and tests are completed;
4. test the system's core modules by implementing unit testing, integration testing, system testing, rule and inference validation, application programming interface communication testing, spatial-query testing, database-operation testing, map interaction testing, DSS decision-flow testing, role-based administrative testing, and end-to-end validation to ensure that susceptibility assessment, geographic selection, dynamic visualization, preparedness guidance, and administrative data management operate accurately and consistently;

5. evaluate the system's overall level of user satisfaction in terms of accessibility, reliability, usability, security, maintainability, and portability of the system using ISO/IEC 25010.

Theoretical Framework

The theoretical framework of FloodSense connects a deterministic Rule-Based Expert System, geographic visualization, and a separate Decision Support System (DSS) to provide scenario-based flood-susceptibility assessment and pre-event preparedness guidance for Bacoar City. The Flutter application sends user-confirmed rainfall and location inputs to Django services backed by PostgreSQL/PostGIS. The framework identifies how controlled knowledge, inference, map presentation, and guidance relate; conceptual components in the architecture are not necessarily operational in the current prototype.

Authentication and Onboarding Module. The architecture reserves a place for resident identity and introductory safety information. Resident registration, login, password recovery, and profile management are not complete user-facing functions; the current assessment flow does not require an account. Any future onboarding should distinguish hypothetical susceptibility information from live forecasts, official warnings, and evacuation orders.

Rainfall and Location Input Module. The resident chooses Light, Moderate, Heavy, Intense, or Torrential rainfall and a duration of 1, 3, 6, 12, or 24 hours, confirms the hypothetical scenario, and explicitly requests an assessment. Location may be selected manually, adjusted with a map pin, or obtained once through foreground device permission; the proposed barangay requires confirmation or correction. The rainfall labels are interface categories, not verified live measurements, and operational thresholds require authoritative confirmation.

Web Administration and Knowledge Management Module. The custom Admin portal supports authenticated dashboard access, read-only map-data and

rainfall-reference review, parameter-governance views, and controlled DSS-content, evacuation-center, data-source, and audit workflows. Ordinary administrators cannot edit raw Expert System rules, priorities, conflict handling, or algorithm logic; technical Django Admin access is separate. A fixed-schema CSV exchange for permitted parameter values is a confirmed requirement, but its export, import, validation, interface, and tests are not yet implemented. Institutional approval and record provenance must be established before data are treated as authoritative.

Django REST and GeoDjango Processing Module. Django and Django REST Framework connect the mobile and administrative interfaces to server-side validation, permissions, assessment, and data services. GeoDjango and PostGIS support polygon and point operations, including proposed barangay detection from a selected coordinate. The mobile application and custom Admin portal do not connect directly to the database.

Rule-Based Expert System Assessment Module. A rule-based system stores domain facts and applies an inference procedure to reach a conclusion (Buchanan & Shortliffe, 1984). FloodSense deterministically matches the confirmed rainfall scenario and supported area against applicable rules, resolves matches and conflicts, and returns a susceptibility class with an explanation. The prototype demonstrates Low, Moderate, High, and Very High outcomes, but its provisional inputs and reference geography do not constitute an approved operational Bacoor City susceptibility dataset.

Dynamic GIS Map and Check My Area Module. The resident map supports panning, zooming, manual selection and pin adjustment, scenario-dependent demonstration coloring, and a neutral 47-barangay administrative reference layer. The layer represents barangay identities through a documented derivation; geometry checks do not make it City-issued or a flood-hazard layer. The optional foreground location request is one-time, not background tracking. Evacuation-center markers and

verified operational susceptibility maps must not be inferred from the current reference display.

Explainable Result Module. The result reports the selected scenario, resolved area, susceptibility class, and the basis supplied by deterministic rule matching, subject to available source and validation information. The same inputs and active rules should yield the same class. An assessment is not a live flood prediction or official warning, and unsupported or unvalidated data must be identified rather than presented as confirmed local risk.

Decision Support System Module. Decision support organizes information and guidance for human judgment (Power, 2002). FloodSense keeps published pre-event preparedness guidance separate from the susceptibility result: guidance may use the result as context but cannot change its class or issue an evacuation order. The current implementation exposes published guidance; a complete resident-facing multi-question decision tree remains unverified.

Evacuation Center Resource Module. The backend and custom Admin portal maintain evacuation-center records. Resident-facing nearby-center discovery, distance ranking, live capacity, operating status, and safe-road routing have not been verified as implemented. Any future public display will require authorized, current records and directions to confirm emergency information with the responsible authorities.

User Account and Preferences Module. Resident account and preference functions remain conceptual rather than completed mobile functions. The current location flow uses a coordinate for the active request and does not continuously track the resident or establish a resident assessment history. Future account features, if implemented, must remain separate from administrative authority over system knowledge and data.

PostgreSQL and PostGIS Database Module. PostgreSQL/PostGIS stores relational and geographic records used by the backend, including reference

boundaries, rainfall references, rule and DSS records, and administrative data. Database models alone do not establish that every corresponding resident function is available. Source status and validation are necessary before provisional or derived records can support public operational claims.

Together, these components explain how confirmed scenario and area inputs pass through controlled services to deterministic susceptibility inference, map display, and separate preparedness guidance. FloodSense is intended to support pre-event understanding without replacing PAGASA, BDRRMO, barangay officials, or emergency responders.

The System Architecture diagram organizes resident and administrator interactions, server-side modules, external information sources, and PostgreSQL/PostGIS storage. It represents the framework's functional relationships; modules shown there must be read with the implementation and data limitations described in the study.

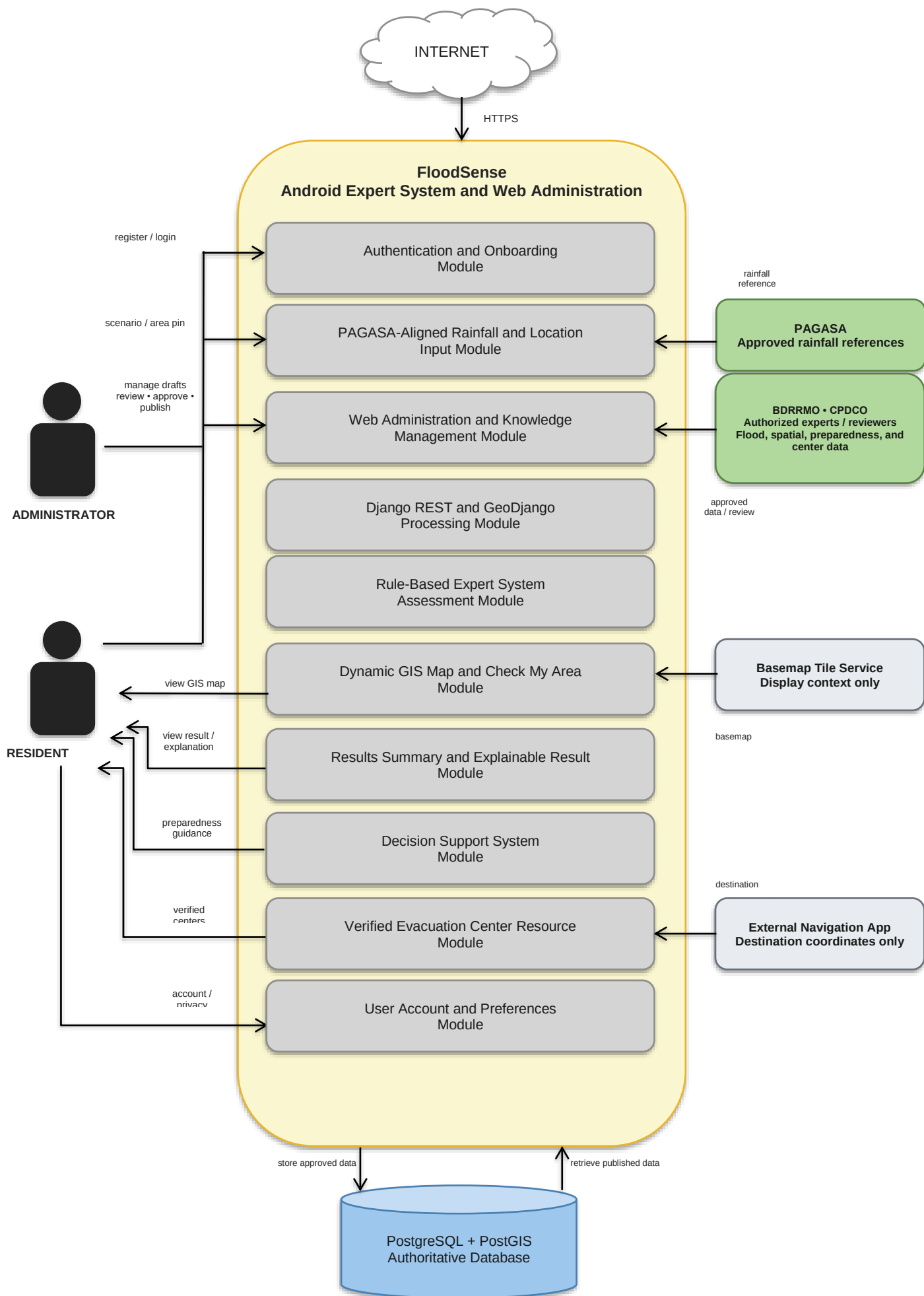


Figure 1. System Architecture of FloodSense.

Significance of the Study

FloodSense investigates how scenario-based susceptibility information and separate preparedness guidance may be made understandable to Bacoor City residents. The following groups may benefit, subject to source validation and completion of evaluation.

For the Bacoor Disaster Risk Reduction and Management Office (BDRMO). The prototype offers a possible channel for communicating validated susceptibility explanations and pre-event guidance. Its custom Admin portal supports controlled content and record workflows, while ordinary administrators cannot rewrite the inference rules. The system supplements, but does not replace, official warnings or emergency directives.

For the City Planning and Development Coordinator Office - Bacoor (CPDCO). The study illustrates how appropriately sourced planning and geographic information could inform location-specific public explanations. City-held maps and datasets require permission, provenance, and technical review before operational use; the current administrative reference boundary is not an official flood-hazard dataset.

For residents of historically flood-affected barangays. Preliminary survey responses point to difficulties with timely, clear, location-specific flood information, although the surveys are not representative of all Bacoor residents. The prototype lets users confirm a hypothetical rainfall scenario and location, request an explained susceptibility assessment, view a demonstration map, and read pre-event guidance. It does not provide official forecasts, warnings, nearby-center discovery, or safe-route advice.

For future researchers. The study documents an artifact-centered approach to deterministic inference, geographic processing, data provenance, and separate preparedness guidance. Its development and eventual evaluation may inform comparable projects, but local rules, spatial data, and findings must be validated anew before transfer to another community.

Time and Place of the Study

The FloodSense study is scheduled for September 2026 to May 2027, with preliminary needs assessment completed before that period. Cavite State University - Bacoor City Campus in Soldiers Hills IV, Molino, Bacoor City, Cavite serves as the academic base for development, consultation, testing, and manuscript preparation. The research follows Design Science Research. The prototype integrates backend and mobile components with automated test suites, while formal resident and system-quality evaluation remains pending.

The first preliminary resident survey involved 50 respondents from 12 Bacoor City barangays and was conducted from May 5 to 12, 2026 using the survey report's purposive-convenience approach. A separate second report dated May 2026 summarized 40 respondents; it does not document precise collection dates, a respondent-selection procedure, or whether participants overlapped with the first survey. These reports are separate needs-assessment evidence, not final software-evaluation results. BDRRMO interviews were recorded at the City of Bacoor Legislative and Disaster Resilience Building, Barangay Bayanan, on May 12 and May 24, 2026; a CPDCO interview was recorded at the Bacoor Government Center, Barangay Bayanan, on May 18, 2026.

The preliminary surveys and interviews informed the problem definition, data requirements, and safety boundaries of the prototype. Prototype design and implementation are ongoing, with automated tests available for core components. Institutional approval of operational susceptibility data, expert validation, and formal resident or system-quality evaluation remain pending. Findings from the new survey and final evaluation will be reported only after their instruments, responses, and analysis are available.

Scope and Limitation of the Study

The study develops FloodSense as a scenario-based, pre-event flood-susceptibility assessment and preparedness-support system for Bacoar City. Its deterministic Rule-Based Expert System evaluates a user-confirmed rainfall scenario and supported location, while the separate Decision Support System presents published preparedness guidance without changing the susceptibility result. The current prototype produces Low, Moderate, High, or Very High demonstration outputs and explanations. Operational use remains blocked until approved local susceptibility data, validated parameters, and institutional authorization are available. FloodSense does not provide real-time flood monitoring, official forecasts, warnings, evacuation orders, or safety guarantees.

The primary intended users of the Android application are Bacoar City residents. In the current mobile flow, a resident selects Light, Moderate, Heavy, Intense, or Torrential rainfall and a duration of 1, 3, 6, 12, or 24 hours, confirms the hypothetical scenario, selects or confirms a location, and explicitly requests an assessment. Resident registration, login, persistent profiles, and assessment history are not exposed in the current interface and remain unimplemented for the intended user flow.

The interactive map is limited to supported Bacoar City reference and demonstration data. It can pan and zoom, display scenario-dependent demonstration colors, accept manual area selection, and show a neutral 47-barangay administrative reference boundary. The boundary layer represents current barangay identities through a documented derivation and has passed technical geometry checks, but it is not automatically City-issued or City-verified and contains no flood-susceptibility facts. The map does not display live flood extent, water depth, traffic conditions, road closures, river level, center occupancy, or satellite rainfall imagery.

Location functions allow the user to select an area manually, place or adjust a map pin, or request the device's location once while the application is in the foreground.

The application can propose a barangay from the coordinate and requires the user to confirm or correct it before assessment. The coordinate is used for the active request and is not continuously collected in the background. The current function does not find safe routes, verify road passability, or continuously track the resident.

The backend and custom Admin portal include an evacuation-center record workflow. This capability does not yet expose resident-facing nearby-center discovery, distance ranking, live capacity, operating status, or safe-route guidance. Any future resident display must rely on authorized and current records and must direct users to confirm emergency instructions through official channels.

The Decision Support System is confined to pre-event preparedness and remains separate from the susceptibility assessment. The current implementation can return published guidance associated with the assessment context. A resident-facing structured multi-question decision tree has not been verified as complete. The DSS does not alter the susceptibility class, issue autonomous evacuation orders, diagnose conditions, guarantee safety, or make the resident's final decision.

A separate custom Web Administration System provides authenticated access to a dashboard, read-only map-data and rainfall-reference review, parameter-governance views, and workflows for DSS content, evacuation-center records, data sources, and audit history. Ordinary administrators cannot directly edit raw Expert System rules, rule conditions, priorities, conflict-resolution logic, or the algorithm. Technical Django Admin remains a separate developer or research-maintenance interface. The confirmed fixed-schema CSV export-and-import workflow is not yet operational and will require routes, services, schema and value validation, interface controls, permissions, integration, and tests before it can be described as implemented.

The implementation stack uses Flutter and Dart for the mobile application; flutter_map for resident-facing mapping; Python Django and Django REST Framework for backend services and REST interfaces; GeoDjango for geographic operations;

PostgreSQL with PostGIS for relational and spatial records; and Django templates with Leaflet for the custom administrative interface. The current prototype requires network access for integrated server functions. iOS support, fully offline operation, live sensors, automated meteorological feeds, background polling, background GPS tracking, emergency dispatch, and machine-learning prediction are outside the current scope.

The reliability of any susceptibility assessment depends on the completeness, geographic resolution, provenance, institutional approval, and expert validation of the data and knowledge used. No approved operational susceptibility dataset is currently present in the repository. The available 47-barangay layer is an administrative reference only, while provisional research materials remain unvalidated or restricted. Demonstration results must therefore not be presented as official Bacoar City flood-hazard information.

Data privacy and security are addressed in accordance with Republic Act No. 10173, or the Data Privacy Act of 2012. The current location flow requests foreground permission only when the user chooses the location function, uses the coordinate for the active request, permits confirmation or correction of the detected barangay, and does not implement continuous background location tracking. Administrative functions require authenticated and authorized access. Evaluation findings will not be generalized until valid survey and system-evaluation data are available.

Definition of Terms

The following terms are defined operationally as they are used in this study to ensure clarity and consistency throughout the research:

Android Application. The resident-facing FloodSense software developed with Flutter. Its current interface supports scenario selection and confirmation, manual or one-time foreground location input, barangay confirmation or correction, assessment requests, map presentation, susceptibility results, explanations, and

preparedness guidance. Resident accounts, assessment history, notifications, and offline operation are not currently exposed.

Check My Area. The location-selection flow that allows manual area selection, placement or adjustment of a map pin, and an optional one-time foreground location request. A detected barangay is proposed for user confirmation or correction before assessment; the function does not continuously track the device.

Color-Coded Susceptibility Levels. The visual scheme used on the dynamic map in which green indicates Low, yellow indicates Moderate, orange indicates High, and red indicates Very High flood susceptibility for the active scenario.

CPDCO (City Planning and Development Coordinator Office). The Bacoor City office that maintains planning and geographic information, including elevation, land-use, flood-hazard, and barangay-boundary materials relevant to the study.

Decision Support System (DSS). The component separate from the Rule-Based Expert System that presents published pre-event preparedness guidance using the assessment context. It does not change the susceptibility result, issue an official warning or evacuation order, or make the user's final decision.

Dynamic Map. The interactive map that supports panning, zooming, scenario-dependent demonstration coloring, manual area selection, map-pin placement, and display of the neutral 47-barangay reference layer. The reference boundary is not an official flood-hazard dataset.

Expert-Provided and Expert-Validated Knowledge. Flood-related facts, parameters, rules, preparedness guidance, and supporting information that have been supplied or reviewed by an authorized source and recorded with adequate provenance. Repository presence or technical validation alone does not establish institutional approval.

Expert System. A knowledge-based computer system that applies stored facts and conditional rules through an inference process. In FloodSense, it produces an explainable susceptibility class for a supported geographic area and rainfall scenario.

Flood Susceptibility. The relative potential of a supported geographic area to experience rainfall-related flooding under a user-confirmed scenario. The current prototype demonstrates Low, Moderate, High, or Very High outputs but does not yet contain an approved operational Bacoar City susceptibility dataset and is not a real-time prediction.

Geographic Information System (GIS). A system for organizing, analyzing, and presenting information associated with geographic locations. FloodSense uses GIS functions for polygons, points, containment queries, interactive visualization, and evacuation-center locations.

GeoDjango. The geographic extension of Django used by FloodSense to represent spatial data and perform geographic operations between backend services and the PostGIS database.

Inference Process. The procedure that matches facts from the selected rainfall scenario and geographic area against validated rules in the knowledge base to derive a susceptibility conclusion and supporting explanation.

Knowledge Base. The structured collection of expert-validated facts, conditional rules, geographic attributes, rainfall parameters, susceptibility categories, and related provenance used by the Expert System.

PAGASA-Aligned Rainfall Input. The current interface labels for Light, Moderate, Heavy, Intense, and Torrential rainfall and durations of 1, 3, 6, 12, or 24 hours. These selections are hypothetical scenario inputs. Their operational thresholds and use require authoritative source confirmation and validation before official deployment.

PostGIS. The spatial extension of PostgreSQL used to store and query FloodSense geographic polygons, points, boundaries, selected locations, and evacuation-center coordinates.

Preparedness Provenance. Information identifying the source, validation status, responsible actor, relevant date, and version of a preparedness item or related

record. Provenance permits users and administrators to distinguish approved, provisional, and unsupported information.

Rainfall Parameters. The supported combination of rainfall intensity and duration selected by the user and applied as scenario facts during Expert System assessment.

Rule-Based Expert System. An Expert System in which domain knowledge is represented as conditional rules. FloodSense evaluates applicable rules against known scenario and geographic facts to produce a susceptibility assessment.

Web Administration System. The custom browser-based interface used by authorized administrators for the dashboard, read-only map and rainfall-reference review, parameter governance, DSS-content management, evacuation-center management, data-source management, and audit history. It does not expose raw rule editing. The separate technical Django Admin is reserved for developer or research maintenance. Fixed-schema CSV export and import is a confirmed but currently unimplemented requirement.

REVIEW OF RELATED LITERATURE

This chapter presents a review of related literature, articles, and studies that provide guidance in identifying and formulating the requirements needed to start the project. Moreover, this chapter was designed to examine relevant concepts, methods, challenges, and practices associated with flood-susceptibility assessment, geographic information systems, Rule-Based Expert Systems, and Decision Support Systems.

Related Foreign Literature

Flood Susceptibility and Scenario-Based Assessment

Flood susceptibility expresses the relative tendency of a location to experience flooding when relevant conditions are present. It differs from flood hazard, which concerns the probability and physical characteristics of an event; exposure, which concerns people or assets in affected places; vulnerability, which concerns the capacity to be harmed; and risk, which combines hazard and potential consequences. Membele, Naidu, and Mutanga (2022) found that flood-vulnerability mapping in developing countries varies substantially in terminology, indicators, spatial scale, data availability, and validation practice. Their review shows why a public system must name the type of assessment it performs and avoid presenting a general vulnerability or hazard map as a locally validated susceptibility conclusion.

Scenario-based assessment supports planning by holding selected conditions explicit instead of implying that the system is observing an event as it occurs. A hypothetical rainfall intensity and duration can be treated as assessment inputs only when their meanings, permitted combinations, and geographic applicability are documented. This supports the FloodSense interaction in which the resident selects or confirms a scenario before requesting assessment. It does not establish that the scenario is current, predicted, or officially issued, and it does not justify importing weights or thresholds derived for another place.

Deterministic Rule-Based Expert Systems and Knowledge Bases

A Rule-Based Expert System represents domain knowledge through facts, conditions, and production rules and applies an inference procedure to derive a conclusion. Buchanan and Shortliffe (1984) provide the foundational distinction between the knowledge base and the inference mechanism. More recently, Papadopoulos et al. (2022) reviewed technologies and standards used in rule-based decision-support systems and described how encoded knowledge and rules engines evaluate case-specific facts. Although their review concerns clinical systems, the architectural distinction is relevant to FloodSense: flood knowledge, scenario facts, geographic facts, and explanations belong to controlled knowledge structures, while the inference procedure determines which eligible rule applies.

Deterministic processing means that the same approved facts and the same published rule version should produce the same result. It does not imply that the rules are scientifically valid merely because the software executes them consistently. Knowledge acquisition, conflict handling, parameter definitions, version control, and expert or institutional validation remain necessary. Accordingly, ordinary FloodSense administrators may manage approved records and permitted values but do not rewrite raw rule conditions, priorities, conflict-resolution logic, or algorithm code. The confirmed fixed-schema CSV exchange is a future controlled-value workflow and is not evidence that such exchange is currently operational.

Explainable Inference

Explainability is necessary when a system produces a classification that may influence preparedness decisions. Kostopoulos, Davrazos, and Kotsiantis (2024) reviewed explainable decision-support approaches and emphasized transparency, interpretability, and human-understandable justification. Their review primarily addresses artificial-intelligence-based systems, but its communication principles apply to deterministic inference: a user should be able to understand the relevant inputs, the

basis of the result, and the limitations of the evidence without being required to inspect source code or raw rule definitions.

For FloodSense, an explanation should identify the confirmed rainfall scenario, resolved geographic context, susceptibility class, applicable basis, source or version information, and any limitation state. This supports accountability while preserving the distinction between explanation and rule administration. It also prevents a color or label from appearing authoritative without disclosing what it represents. Explainability does not remove the need for validated data, and a clear explanation cannot turn provisional information into an official local assessment.

Decision Support and Preparedness Guidance

A Decision Support System organizes information, alternatives, or recommended actions to assist human judgment. In flood preparedness, support may include understandable mitigation choices, household preparation, communication planning, protection of documents and valuables, assistance for vulnerable household members, attention to official advisories, and evacuation readiness. Mostafiz et al. (2022) observed that many web-based flood tools provide technical information but do not consistently provide locally actionable information for individuals and communities. Their review supports combining understandable risk communication with practical guidance while clearly stating the tool's authority and limitations.

FloodSense separates the susceptibility assessment from preparedness guidance. The deterministic Expert System produces the assessment, while the DSS presents source-backed pre-event guidance without changing the susceptibility class. Information about evacuation facilities may support readiness when its source, verification date, location, and operational limitations are known, but distance alone does not prove route safety, current opening, accessibility, or available capacity. The DSS therefore assists preparation and does not issue an official warning, evacuation order, or guarantee of safety.

GIS, Interactive Maps, and Spatial Databases

Geographic Information Systems organize, analyze, and present information associated with locations. Their value in flood research depends on the suitability of the layers, coordinate reference systems, spatial resolution, classification methods, and validation procedures. An administrative boundary can resolve the name of a barangay or provide map context, but it contains no flood-susceptibility fact unless appropriate hazard or susceptibility evidence has been associated with it. This distinction is important when users may interpret any colored polygon as an official hazard layer.

Spatial databases support the controlled storage and querying of geographic records. The PostGIS Development Group (2026) explains that PostGIS extends PostgreSQL with geometry and geography types, spatial indexes, spatial relationships, coordinate transformation, validity operations, and spatial processing. These capabilities are relevant to point-in-polygon queries, administrative-area lookup, map-feature storage, and provenance-linked geographic records. They provide technical infrastructure, not scientific validation: a valid geometry and a successful spatial query do not establish that a flood conclusion is institutionally approved.

Flood-Risk Communication

Flood-risk information must be understandable, actionable, and explicit about uncertainty. Mostafiz et al. (2022) identified gaps in web-based flood communication, particularly where tools provide information without sufficiently connecting it to decisions that residents can make. Clear terms, legends, labels, explanation text, and source notices help prevent non-specialists from treating model outputs as certain predictions. Communication should also distinguish long-term or scenario-based information from time-sensitive official warnings.

These principles support a guided FloodSense flow rather than an unexplained technical map. Susceptibility colors must be paired with text, the selected scenario,

and limitations. Preparedness information must direct users toward official channels during actual events. The interface must not imply that a neutral boundary layer, a demonstration color, or a provisional record represents a current flood condition.

Data Provenance and Geographic Validation

Provenance records where information originated, how it was transformed, which version was used, and what review or approval applies. Tullis and Kar (2021) connected provenance to ethical replicability and reproducibility in GIScience, including critical applications in which geographic data affect public decisions. Fakhruddin et al. (2022) similarly emphasized risk-informed data practices and the importance of understanding how disaster and climate data are produced, shared, interpreted, and governed.

For FloodSense, provenance should accompany susceptibility parameters, geographic layers, preparedness items, and resource information. Technical geometry validation may confirm coordinate-system consistency, valid shapes, or feature counts, but institutional validation concerns authority, currency, meaning, and permitted public use. The two forms of validation are related but not interchangeable. This distinction prevents a technically processed dataset from being represented as official Bacoor City flood information without documented approval.

Foreground Location and Privacy

Mobile location can reveal a person's position and may contribute to identification or tracking. Android Developers (2026) distinguishes foreground access, one-time access, and background access and recommends requesting only the permission needed for the active function. This supports a FloodSense flow in which location is obtained only after a user initiates location selection or assessment, while manual selection remains available when the resident declines permission or prefers another area.

Foreground acquisition does not imply continuous monitoring. FloodSense does not require background GPS tracking, automatic polling, or autonomous location-based alerts. Location handling should remain purpose-limited to the active assessment, use only the precision necessary for the supported geographic operation, and avoid retaining unnecessary movement information. The presentation must also allow the user to confirm or correct the detected barangay because device accuracy and boundary proximity can affect geographic resolution.

Software Testing and Quality Evaluation

Software testing establishes whether specified behavior is implemented under defined conditions, while quality evaluation examines broader product characteristics. ISO/IEC 25010:2023 defines a product-quality model with nine characteristics for specifying and evaluating information and communication technology products (International Organization for Standardization [ISO], 2023). The model can guide the selection of criteria for FloodSense, but it does not provide a ready-made acceptance result or replace an approved evaluation instrument.

A defensible evaluation must identify the tested build, participants or evaluators, tasks, criteria, scale, procedure, valid responses, and interpretation method. Repository tests and interface demonstrations can support implementation verification, but they are not resident-acceptance findings. Because the final FloodSense survey and evaluation are not complete, Chapter 2 uses ISO/IEC 25010 only as literature supporting future assessment and does not report quality scores.

Related Local Literature

Philippine Flood Conditions and Institutional Preparedness

Official Philippine statistics continue to record substantial effects from extreme events and disasters. The Philippine Statistics Authority (PSA, 2026) summarizes national information on disaster occurrence, affected populations, deaths, and

economic losses. These national figures establish the continuing relevance of disaster preparedness but do not identify the susceptibility of a particular Bacoar barangay or validate the inputs of a local application.

The National Disaster Risk Reduction and Management Council's National Disaster Response Plan identifies institutional responsibilities and coordinated actions for hydro-meteorological hazards (NDRRMC, 2024). The plan reinforces that warnings, response decisions, and emergency coordination belong to authorized government mechanisms. FloodSense may support pre-event understanding and preparedness, but residents must continue to follow PAGASA, local disaster offices, barangay authorities, and emergency responders during an actual event.

Official Hazard Information and Warning Authority

PAGASA operates rainfall and flood-information processes that use observations, forecasts, water-level information, and coordinated warning protocols. Its 2023 Annual Climate Bulletin explains the role of rainfall warning systems and the use of official warning levels for public action (PAGASA, 2024). FloodSense rainfall selections are hypothetical assessment inputs; they are not automatic updates from PAGASA, an interpretation of current rainfall, or an official warning product.

GeoRisk Philippines presents HazardHunterPH as a national platform for indicative hazard assessment while advising users that results depend on available government data and do not replace official hazard reports from mandated agencies (GeoRisk Philippines, n.d.). Its source notices and limitations demonstrate responsible hazard communication. The same principle applies to FloodSense: sources, dates, limitations, and validation status should remain visible, and an indicative or provisional record should not be represented as an absolute prediction.

Cavite, Bacoar City, and the Imus River Basin

The Japan International Cooperation Agency reported the inauguration of flood-mitigation facilities associated with the Imus River Basin and low-lying areas of Imus and Bacoar (JICA, 2021). This institutional record confirms the relevance of watershed and infrastructure conditions to the local context. It does not provide barangay susceptibility classifications, rainfall-rule thresholds, or validation for FloodSense, and it should not be treated as a substitute for hydrologic or hazard data.

Administrative geography in Bacoar has also changed. The PSA documented the 2023 merger and renaming of barangays and currently lists 47 barangays in the City of Bacoar (PSA, 2023, 2025). These records support the current identities used in a reference layer. They do not establish the accuracy of every derived polygon, City verification of a third-party boundary, or flood conditions within any barangay. Administrative reference and flood-susceptibility evidence must therefore remain separate.

Administrative Data Governance and Responsible Use

Government publications, local records, and research datasets differ in authority and permitted use. An official list of administrative units may verify names and codes, while an authorized hazard agency may provide hazard information, and a local office may validate operational records. Combining these sources requires documentation of ownership, date, scale, processing, approval, and restrictions. Provenance is particularly important when a dataset has been transformed, because technical processing can change geometry or attributes without changing the institution responsible for the original information.

FloodSense consequently distinguishes approved, provisional, technically validated, and unsupported information. The 47-barangay layer may serve as a current administrative reference after documented derivation and geometry checks, but it is not a flood-susceptibility layer and is not automatically City-issued or City-verified. No

operational susceptibility dataset should be described as established until the necessary source, method, expert review, institutional authority, and database integration are documented.

Philippine Data Privacy and Software-Quality Context

Republic Act No. 10173, the Data Privacy Act of 2012, requires personal-information processing to observe transparency, legitimate purpose, and proportionality. For a resident-facing location function, these principles support clear notice, user initiation, limited collection, secure handling, and avoidance of continuous background tracking. They also support role-controlled administrative access. The Act is legally authoritative despite falling outside the preferred publication period for recent literature.

Philippine software studies have also used ISO/IEC 25010 as an evaluation framework, but the criteria and edition must be identified accurately. An evaluation of another application cannot be transferred to FloodSense, and the presence of a questionnaire does not establish completed results. FloodSense quality conclusions will require its own approved instrument, respondents or evaluators, documented procedure, and completed analysis.

Related Foreign Studies

FloodAdapt: An Open-Source Application for Flood-Risk Scenario Modeling and Adaptation Planning

Deltares (2025) developed FloodAdapt as an open-source decision-support application for exploring flood risk under alternative present and future conditions. The application links the SFINCS flood model with Delft-FIAT impact assessment and allows users to define what-if events, future scenarios, and adaptation measures. Its outputs include mapped flooding and impacts, summary metrics, and cost-benefit

information intended for planners, analysts, and decision makers rather than for automatic public warning.

FloodAdapt shows how explicit scenarios can be compared without presenting them as observations of a current event. Its physics-based modeling, impact calculations, and adaptation analysis require specialized data and expertise, and the downloadable release does not supply Bacoar-specific inputs or institutional approval. The model configuration, damage functions, and scenario assumptions therefore cannot be transferred directly to Bacoar City.

FloodAdapt is similar to FloodSense in asking users to consider a defined scenario and in presenting mapped decision information. It differs in audience, scale, scientific method, and output: FloodSense uses a resident-confirmed rainfall scenario and deterministic rules to produce a bounded susceptibility class followed by separate preparedness guidance. FloodSense does not perform hydrodynamic simulation, damage estimation, adaptation comparison, or cost-benefit analysis.

GloFAS: An Operational Platform for Global Flood Forecasting and Monitoring

The Copernicus Emergency Management Service (2026) operates the Global Flood Awareness System, or GloFAS, as a global flood-forecasting and monitoring service. Its web interface provides map layers, forecast products, hydrological information, and data access intended to support preparedness and emergency-management activity across transnational river basins. The service draws on operational forecasting infrastructure and is maintained within the Copernicus Emergency Management Service and the European Centre for Medium-Range Weather Forecasts.

GloFAS demonstrates the institutional, computational, and data requirements of an operational forecast service. Global model resolution, forecast uncertainty, local river behavior, and the need for national or local interpretation limit direct use for

household-level decisions. Its products also do not constitute Bacoar City validation and cannot be converted into FloodSense rules without an approved methodology, local evidence, and appropriate institutional authority.

Both GloFAS and FloodSense use maps and flood-related categories to support understanding, but their functions are fundamentally different. GloFAS is an operational monitoring and forecasting service; FloodSense is a user-requested, scenario-based pre-event assessment. FloodSense does not ingest live GloFAS forecasts, update rainfall automatically, monitor rivers, or issue official warnings.

IFIS: A Public Web Platform for Real-Time Flood Information, Sensor Data, and Forecasts

The Iowa Flood Center (2024) describes the Iowa Flood Information System, or IFIS, as a publicly accessible platform that integrates a statewide sensor network, weather and hydrologic information, flood-inundation maps, and real-time forecasts. The system has evolved since its 2011 launch and supports residents, emergency managers, researchers, and other users through an interactive geographic interface. Its operation depends on Iowa-specific sensors, models, data streams, and institutional maintenance.

The Iowa Flood Center (n.d.) also states that the platform is under active development and that its maps and information are provisional and are not a substitute for official regulatory determinations. This qualification is important because technically sophisticated public maps can still have geographic, predictive, and legal limitations. IFIS does not provide an approved susceptibility dataset, rule base, or evaluation result for Bacoar City.

IFIS resembles FloodSense in offering location-oriented map information and understandable public presentation. It differs because IFIS uses real-time sensor and forecasting infrastructure, whereas FloodSense requires a resident to confirm a hypothetical scenario and explicitly request deterministic assessment. FloodSense has

no statewide sensor network, continuous monitoring, autonomous alerting, or real-time forecast function.

RAPT: A Geographic Information System for Community Resilience Planning and Data Visualization

The Federal Emergency Management Agency (FEMA, 2025) maintains the Resilience Analysis and Planning Tool, or RAPT, as a public geographic application for community resilience planning. The tool brings together more than one hundred layers concerning population, infrastructure, hazards, and community conditions and supports visualization, filtering, comparison, and export for planning, mitigation, response, and recovery activities in the United States.

RAPT illustrates the value of bringing contextual data and provenance into one map, but it is not limited to floods and does not convert every displayed layer into a location-specific hazard conclusion. Its United States datasets, administrative units, definitions, and legal context are not applicable to Bacoor without local substitution and validation. Layer availability also does not guarantee suitability for a particular decision.

RAPT and FloodSense both use geographic interfaces to organize information for decision support. FloodSense differs by applying a protected deterministic inference process to a resident-confirmed rainfall scenario and then presenting separate preparedness guidance. FloodSense does not reproduce RAPT's national data catalog, demographic analysis, resilience indicators, or export capabilities.

Check the Long Term Flood Risk

The Environment Agency (2024) supports England's national assessment of flood and coastal erosion risk, while GOV.UK (n.d.) offers the public Check the Long Term Flood Risk for an Area service, which allows a user to search a place and review long-term risk from rivers and the sea, surface water, reservoirs, and groundwater. The

service also provides contextual information on climate change and links users to planning and preparedness resources through an official government channel.

The service explicitly distinguishes area-level information from the risk of an individual property. That limitation reflects differences in spatial scale, source datasets, update cycles, and the uncertainty of local conditions. Its English hazard models, categories, responsibilities, and official status cannot be generalized to Bacoor City or used to validate FloodSense.

The location-search interaction and careful limitation statements are relevant to FloodSense's map and communication design. FloodSense, however, is not a national long-term-risk authority and does not cover multiple hazard mechanisms or property-level risk. It presents a bounded hypothetical-scenario susceptibility assessment and directs users to official agencies for current forecasts, warnings, and emergency instructions.

Related Local Studies

UP NOAH Know Your Hazards and NOAH Studio: Interactive Philippine Platforms for Hazard Mapping and Flood-Scenario Visualization

The UP NOAH Center (n.d.) maintains public location-based interfaces through Know Your Hazards and NOAH Studio. These platforms allow users to search places, view hazard layers, and examine flood scenarios such as the five-, twenty-five-, and one-hundred-year layers derived from simulations, satellite information, and historical or geographic datasets. Their public maps support awareness and exploration across Philippine locations.

UP NOAH also publishes disclaimers on the interpretation and authority of its layers, including the limits of administrative boundaries and the need to consult appropriate agencies. Return-period flood scenarios are not the same as the rainfall-intensity and duration selections used by FloodSense. The platform's layers, simulations, and classifications cannot be treated as imported or approved

FloodSense data unless provenance, permission, methodology, validation, and integration are separately established.

The UP NOAH interfaces are similar to FloodSense in their use of interactive maps and location-based flood communication. They differ in scenario definition, data production, geographic coverage, and institutional purpose. FloodSense does not claim to reproduce NOAH simulations, predict inundation extent, or provide the same nationwide hazard products.

HazardHunterPH: A Philippine Location-Based Application for Indicative Multi-Hazard Assessment

GeoRisk Philippines (n.d.) presents HazardHunterPH as a public web application that generates indicative multi-hazard assessment reports for a user-selected Philippine location. The platform combines hazard information from mandated government agencies and communicates the available layers, source institutions, and assessment limitations. It is intended to improve public access to location-based hazard information rather than to replace the official reports issued by responsible agencies.

HazardHunterPH's own guidance states that the report is indicative and should not be used as an official or legal document. Available results depend on the datasets supplied to the platform, and a location report does not eliminate the need for agency confirmation. These limitations are directly relevant to Bacoar because neither a national platform nor an administrative reference layer automatically establishes FloodSense's local susceptibility parameters.

HazardHunterPH and FloodSense both use a selected location and a map to communicate hazard-related information. HazardHunterPH retrieves available government hazard layers across multiple hazards, whereas FloodSense applies deterministic rules to a user-confirmed hypothetical rainfall scenario and presents separate preparedness guidance. FloodSense must preserve HazardHunterPH's

lesson on visible sources and limitations without implying that the two systems have the same data, authority, or output.

GeoMapperPH: A Governed Geospatial Application for Collecting Hazard, Exposure, Vulnerability, and Coping-Capacity Data

The Department of Science and Technology-Philippine Institute of Volcanology and Seismology (DOST-PHIVOLCS, 2022a) describes GeoMapperPH as a GeoRisk Philippines application used by trained local-government and agency personnel to collect information on hazards, exposure, vulnerability, and coping capacity. The platform supports standardized field data collection and contributes records to exposure databases that can be used in risk assessment and planning. Its documented training context shows that data collection is governed by authorized users, defined procedures, and institutional coordination.

GeoMapperPH is not presented as an unrestricted resident assessment application. The usefulness of its records depends on collection quality, classification consistency, source responsibility, access controls, and continued updating. Training materials or the existence of a collection application do not prove that its datasets have been approved for FloodSense or imported into the FloodSense database.

The application is relevant to FloodSense because it illustrates provenance, standardized schemas, permissions, and administrative data governance. FloodSense differs in serving a resident scenario-assessment flow and in protecting its deterministic rule logic from ordinary administrators. The confirmed fixed-schema CSV workflow would govern permitted values rather than reproduce GeoMapperPH's field-data collection functions, and that workflow is not yet operational.

PlanSmartPH Ready to Rebuild: A Web-Based Rehabilitation and Recovery Planning Application for Local Government Units

DOST-PHIVOLCS (2022b) describes PlanSmartPH Ready to Rebuild as a web application developed with national agencies and development partners to assist local governments in preparing rehabilitation and recovery plans. The platform organizes baseline data, damage and loss information, recovery needs, programs, and planning outputs so that local officials can develop structured post-disaster plans more efficiently.

PlanSmartPH operates in an institutional planning context and depends on the completeness and validation of local-government data. Its recovery-planning workflow, participating agencies, datasets, and user roles differ from a resident-facing pre-event susceptibility tool. It does not provide approved Bacoor rainfall-rule parameters or establish that FloodSense can make official recovery, response, or evacuation decisions.

Both applications demonstrate the value of structured data and decision support, but they support different phases of disaster management. PlanSmartPH assists post-disaster rehabilitation and recovery planning, while FloodSense supports pre-event scenario understanding and household preparedness. FloodSense therefore does not inherit PlanSmartPH's planning authority, damage assessment, financial estimates, or institutional outputs.

Web-Based Solution for Flood Warning Decision Support in Leyte: A Sensor-Based Flood-Monitoring and Warning Prototype

Bentoso et al. (2021) developed and tested a web-based flood-warning decision-support prototype for the Province of Leyte. The researchers used rainfall and water-level sensors, data communication, a web dashboard, graphical readings, warning thresholds, and notification functions to help responsible users monitor

conditions in flood-prone areas. The conference paper describes a prototype and its functional testing; it does not establish province-wide operational deployment.

The system's sensor locations, communication infrastructure, rainfall and water-level thresholds, and warning procedures were designed for its own context. Those elements cannot be transferred to Bacoar without a new scientific and institutional basis. A working prototype also does not prove the accuracy or authority of warnings outside the study setting.

The Leyte application and FloodSense both organize flood-related information through a digital interface and aim to support preparedness decisions. The difference is decisive: the Leyte prototype monitors sensors and generates warning information, while FloodSense has no continuous monitoring, background polling, autonomous alerts, or official warning authority. FloodSense remains a user-requested hypothetical scenario assessment with separate guidance.

Synthesis of the Reviewed Literature

The reviewed literature separates susceptibility from hazard, exposure, vulnerability, and risk. Membele et al. (2022) show why mapping concepts and validation methods must be identified before a result is communicated. Mostafiz et al. (2022) add that a technically informative map is less useful to residents when it does not connect its result to understandable action. Together, these works support FloodSense's separation of a user-requested susceptibility assessment from source-backed preparedness guidance.

The reviewed systems demonstrate different forms of geographic decision support. Deltares (2025) uses model-based what-if scenarios in FloodAdapt, while the Copernicus Emergency Management Service (2026) and Iowa Flood Center (2024) provide operational forecasts or monitoring information. In the Philippines, the UP NOAH Center (n.d.) and GeoRisk Philippines (n.d.) present location-based hazard

information, whereas DOST-PHIVOLCS (2022a, 2022b) documents agency-oriented data collection and recovery planning. Bentoso et al. (2021) instead investigated sensor-based warning in Leyte. Their data, geographic scale, intended users, and authority differ from those of FloodSense; their thresholds and findings cannot simply be applied to Bacoar City.

Tullis and Kar (2021) emphasize provenance in geographic work, and the Philippine platforms' own source and boundary notices illustrate why technical processing must be distinguished from institutional validation. For FloodSense, a 47-barangay administrative reference layer cannot by itself establish flood susceptibility. Deterministic rule matching may make a result repeatable and explainable, but its public validity still depends on locally appropriate data, documented rules, expert review, and authorized use.

Research Gap

Existing platforms already provide scenario simulation, official hazard-layer lookup, real-time monitoring, forecasts, and institutional planning. The reviewed evidence does not establish that these distinct functions are routinely combined for Bacoar residents in a single flow that starts with a user-confirmed hypothetical rainfall scenario, resolves a supported location, applies deterministic and explainable rules, and then presents separate pre-event preparedness guidance. This is a context-specific design gap, not a claim that FloodSense is the first flood application.

The gap also concerns evidence and authority. Bacoar-specific susceptibility inputs, rule thresholds, geographic layers, and guidance require source documentation and local expert or institutional validation before public conclusions can be treated as operational. FloodSense therefore remains a prototype for scenario-based

assessment and preparedness support, not a substitute for official forecasting, flood warnings, evacuation orders, or verified hazard certification.

Comparative Synthesis of Related Studies

The following comparisons summarize the documented approaches and contributions of the foreign and Philippine systems discussed above. Official platforms are identified by their responsible institutions; the methodology column describes the documented technical approach and does not imply that every platform is a peer-reviewed research study.

Table 1 compares the related foreign systems by responsible institution, documented approach, objective, and contribution.

Table 1. Comparison of related foreign studies

SYSTEM	AUTHOR(S) /YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
FloodAdapt: An Open-Source Application for Flood-Risk Scenario Modeling and Adaptation Planning	Deltares (n.d.)	SFINCS scenario simulation followed by Delft-FIAT impact analysis.	The application aims to help planners examine flooding under selected scenarios. It also aims to compare adaptation measures.	FloodAdapt produces simulated flood and impact maps with summary metrics. Users can compare the outcomes of different scenarios.
GloFAS: An Operational Platform for Global Flood Forecasting and Monitoring	Copernicus Emergency Management Service (2023)	LISFLOOD OS v4 spatial hydrological modeling for river forecasts.	The service aims to provide global river-flood forecasts for early awareness. It makes forecast information available to support preparedness.	GloFAS produces model-derived river-discharge forecasts and mapped products. It makes flood information available across large river basins.

Table 1. Continued

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
IFIS: A Public Web Platform for Real-Time Flood Information, Sensor Data, and Forecasts	Iowa Flood Center (n.d.)	Radar-driven Hillslope Link Model rainfall-runoff forecasting.	IFIS aims to make Iowa flood observations and forecasts accessible to communities. It helps users examine changing conditions through a public map.	The platform brings observations, alerts, forecasts, and flood maps into one interface. Its forecasts draw on Iowa-specific rainfall data and hydrologic modeling.
RAPT: A Geographic Information System for Community Resilience Planning and Data Visualization	Federal Emergency Management Agency (2025)	GIS comparison of hazard, infrastructure, and community layers.	RAPT aims to help emergency managers identify resilience challenges. It supports preparedness, mitigation, response, and recovery planning.	The tool provides geographic layers that users can examine together. These comparisons give planners context about hazards and community conditions.
Check the Long Term Flood Risk	GOV.UK (n.d.)	Computer-modeled area risk across multiple flood sources.	The service aims to help people check long-term flood risk for an area in England. It provides information about flood sources and preparedness resources.	It presents broad area-level flood-risk estimates. Its terms explain that the result cannot predict whether an individual property will flood.

Table 2 compares the related Philippine systems using the same categories.

Table 2. Comparison of related local studies

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
UP NOAH Know Your Hazards and NOAH Studio: Interactive Philippine Platforms for Hazard Mapping and Flood- Scenario Visualization	UP NOAH Center (n.d.)	Point lookup using simulated, satellite, and historical hazard data; flood-map layers.	The interfaces aim to make location-based hazard information accessible to Philippine users. They let users examine a selected point and surrounding mapped hazards.	Know Your Hazards presents point-level hazard information, while NOAH Studio provides flood- hazard layers. UP NOAH also publishes limitations on interpreting its maps.
HazardHunter PH: A Philippine Location- Based Application for Indicative Multi-Hazard Assessment	GeoRisk Philippines (n.d.)	Point lookup of agency hazard layers; exposure by spatial intersection.	HazardHunterP H aims to provide an indicative assessment for a selected Philippine location. It helps users view available multi-hazard information in one report.	The application combines agency-supplied layers in a location-based assessment. Its methods explain calculation assumptions and data limitations.

Table 2. Continued

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
GeoMapperPH: A Governed Geospatial Application for Collecting Hazard, Exposure, Vulnerability, and Coping-Capacity Data	Soriano (2022)	Web/mobile collection of hazard and exposure data into GeoRiskPH.	GeoMapperPH aims to help agencies and local governments gather hazard, exposure, vulnerability, and coping-capacity data. It supports the development of geospatial exposure records.	The official training report documents web and mobile data collection for local-government use. It also reports that General Luna, Quezon received an LGU account and database dashboard.
PlanSmartPH Ready to Rebuild: A Web-Based Rehabilitation and Recovery Planning Application for Local Government Units	Soriano (2022)	Baseline-data assessment and pre/post-disaster recovery-plan inputs.	The application aims to help local governments formulate rehabilitation and recovery plans. It organizes information needed by technical personnel for post-disaster planning.	The official training report documents hands-on entry of disaster-related data. It presents the application as an aid for preparing structured recovery plans.

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Web-Based Solution for Flood Warning Decision Support in Leyte: A Sensor-Based Flood- Monitoring and Warning Prototype	Bentoso et al. (2021)	Rain/water-level sensing, web visualization, and alert messaging.	The researchers aimed to develop a web-based aid for monitoring flood conditions in Leyte. They intended its readings and alerts to support warning decisions.	The authors report graphical presentation of flood conditions. They also report alert messages received by authorities.

[illegible]

Legend:

✓ - included in the documented system; blank - not established by the inspected source.

S1 - FloodAdapt: An Open-Source Application for Flood-Risk Scenario Modeling and Adaptation Planning

S2 - GloFAS: An Operational Platform for Global Flood Forecasting and Monitoring

S3 - IFIS: A Public Web Platform for Real-Time Flood Information, Sensor Data, and Forecasts

S4 - RAPT: A Geographic Information System for Community Resilience Planning and Data Visualization

S5 - Check the long term flood

S6 - UP NOAH Know Your Hazards and NOAH Studio: Interactive Philippine Platforms for Hazard Mapping and Flood-Scenario Visualization

S7 - HazardHunterPH: A Philippine Location-Based Application for Indicative Multi-Hazard Assessment

S8 - GeoMapperPH: A Governed Geospatial Application for Collecting Hazard, Exposure, Vulnerability, and Coping-Capacity Data

S9 - PlanSmartPH Ready to Rebuild: A Web-Based Rehabilitation and Recovery Planning Application for Local Government Units

S10 - Web-Based Solution for Flood Warning Decision Support in Leyte: A Sensor-Based Flood-Monitoring and Warning Prototype

METHODOLOGY

This chapter explains the materials, research design, development process, participants, and evidence-gathering and evaluation procedures used for FloodSense. It distinguishes the work already documented in the prototype from validation and evaluation activities that remain to be completed.

Materials

The software materials are Flutter and Dart for the Android interface; Python, Django, Django REST Framework, and GeoDjango for services and geographic operations; PostgreSQL/PostGIS for relational and spatial storage; and Django templates with Leaflet for the custom administrative portal. Version-controlled source code, migrations, automated backend and mobile tests, interface designs, and research instruments support the work.

Research materials include the two preliminary resident-survey reports, their available questionnaire and response appendix, recorded BDRMO and CPDCO interviews, development documentation, and the 47-barangay administrative reference layer. The boundary layer supports geographic identification but contains no susceptibility facts and is not City-verified. Other provisional geographic material and processing outputs cannot be treated as approved operational data without provenance, authorization, and validation.

Method

The study uses Design Science Research as its research design because its central artifact is a mobile and web system intended to address an identified information problem. Its stages are problem identification, solution objectives, design and development, demonstration, evaluation, and communication (Peppers et al., 2007). An iterative and incremental approach guides software development: the prototype integrates backend, mobile, geographic, and administrative components,

while test and evaluation findings may inform subsequent increments. Formal resident evaluation and any resulting redesign remain pending.



Figure 2. FloodSense Design Science Research and iterative development method (researchers' design, 2026).

The figure shows the six research stages. Its return arrow represents the intended route from evaluation findings to another design-and-development increment; it does not imply that final resident evaluation or a resulting rebuild has already occurred. The following sections identify the input, output, and current completion state of each stage.

System Development Approach and Checkpoints

The prototype comprises backend assessment services, separate DSS guidance, REST endpoints, a Flutter assessment interface, an interactive demonstration map, one-time foreground location with barangay confirmation, and selected administrative workflows. Integrated source code and automated test cases cover these components. Formal resident evaluation and evaluation-driven redesign remain pending.

Administrative design separates ordinary content management from technical rule maintenance. Raw rule editing is not available in the custom Admin portal. Fixed-schema CSV exchange for permitted parameter values remains a requirement for

future implementation and is not operational. Subsequent prototype changes will be assessed through testing and formal stakeholder evaluation.

Problem Identification and Solution Objectives

Problem identification was completed provisionally using the first resident survey of 50 respondents conducted May 5–12, 2026, a separate 40-respondent report dated May 2026, and recorded BDRMO and CPDCO interviews. In the first survey, 46 of 50 respondents reported flooding at least occasionally, 10 of 50 considered themselves well-informed about flood-prone areas, and 32 of 50 reported a wrong decision associated with insufficient flood information. In the second report, 31 of 40 said their area flooded almost every rainy season, 9 of 40 considered themselves well-informed, and 26 of 40 selected one of the two dissatisfied categories for existing flood information. The two instruments and samples are not interchangeable, and neither supports a Bacoor-wide prevalence estimate. The stage output was a preliminary problem definition and set of information needs.

The solution-objectives stage converted these needs and institutional constraints into the existing study objectives: a user-confirmed rainfall scenario, supported area selection, deterministic and explainable susceptibility assessment, geographic presentation, and separate pre-event guidance. The initial objective definition is complete, but data-dependent details remain subject to institutional and expert validation. Additional maps and datasets require access, provenance, institutional permission, and validation before operational use. The scope excludes live forecasts, background monitoring, official warnings, autonomous alerts, and automatic evacuation orders.

Design and Development

Design and development are ongoing. The integrated prototype includes the domain model and database, REST services, deterministic inference and DSS logic, mobile assessment flow, interactive demonstration map, one-time foreground location

with barangay confirmation, and selected custom Admin workflows. The mobile interface requests scenario options and geographic data from backend services and submits selected inputs for assessment. The prototype demonstrates these functions with provisional or demonstration data. Institutional approval of an operational Bacoor City susceptibility dataset, the fixed-schema CSV exchange, resident accounts, resident-facing nearest-center discovery, and production deployment remain outstanding.

The Expert System evaluates supported geographic and rainfall facts through deterministic rules; its susceptibility output is not changed by DSS content. The DSS uses that result as context for published preparedness guidance. Administrative records, geographic processing, validation, provenance, and permissions support these flows, but ordinary administrators cannot directly alter raw rule conditions, priorities, conflict handling, or code. Fixed-schema CSV export and import for permitted values remain a confirmed but unimplemented requirement. Technical Django Admin access for developer or research maintenance is not the ordinary custom Admin portal.

Demonstration

The prototype supports a controlled demonstration with fictional areas: the user confirms a hypothetical rainfall intensity and duration, selects or confirms a location, explicitly requests an assessment, and reviews the explanation, map display, and available guidance. These functions demonstrate software behavior only; they do not validate real Bacoor susceptibility classifications or constitute final stakeholder evaluation. A demonstration using approved local data remains contingent on authorization and validation.

Evaluation and Iteration

Evaluation is partly prepared and partly planned. Backend and mobile automated test cases address core inference, validation, API responses, map behavior, location handling, and administrative access. Their presence does not

constitute a completed resident usability study or final thesis evaluation. Further testing will verify rule matching and conflicts, scenario and geographic validation, API and mobile integration, map behavior, permissions, privacy, and separation of assessment from DSS guidance. Defects or validated stakeholder findings will feed a new design-and-development increment followed by retesting.

ISO/IEC 25010:2023 will guide the planned product-quality evaluation, using characteristics relevant to the approved FloodSense instrument. It is a software product-quality model, not the research design or SDLC. Task-based review of the resident and Admin workflows may supplement technical checks. Participant selection, finalized instruments, responses, scores, and conclusions will be documented only after the new survey and evaluation are completed; the preliminary needs surveys cannot substitute for them.

Communication

Communication is ongoing through the manuscript and development records. The final study will report the implemented artifact, validated sources, completed tests, evaluation findings, and unresolved limitations. Claims about system quality, public utility, or Bacoar-wide applicability will depend on the corresponding evidence and local authorization.

Participants of the Study

The first preliminary survey report describes 50 Bacoar City resident respondents across 12 barangays, gathered May 5–12, 2026 through purposive-convenience selection. Its accompanying appendix contains the questionnaire and response summaries. A second, separate survey analysis report dated May 2026 covers 40 respondents identified with Bacoar barangay labels, but does not establish exact collection dates, how respondents were selected, whether any person also answered the first survey, or the source of every compiled response. Accordingly, the groups are described separately and their counts and percentages are not pooled.

Key informants included a BDRMO staff member interviewed on May 12 and May 24, 2026, and the CPDCO Officer-in-Charge interviewed on May 18, 2026. Their statements informed institutional workflow, data-availability, provenance, and public-communication requirements; they were not a completed test of the FloodSense application. The participants and results of the new resident survey and final system evaluation are not yet available and are not reported here.

Data Gathering and Analysis

The preliminary questionnaires asked about flood experience, information access, interpretation, and desired preparedness support. The interview transcripts were examined for reported office practices, data formats, access restrictions, and concerns about susceptibility interpretation. Development evidence was checked against integrated code and available tests to separate operating features from models, plans, diagrams, and proposals. Survey reports provide descriptive frequencies and percentages only for their own respondents; no combined sample, causal relationship, or citywide estimate is inferred.

Institutional documents and geographic layers are recorded by source and approval status. The 47-barangay reference boundary may support selection and display after technical geometry checks, but those checks do not establish City verification or flood susceptibility. Provisional materials, unavailable operational datasets, and any unapproved inferred rules will not be represented as authoritative Bacoor City information.

System Evaluation

The planned evaluation will assess whether the prototype performs its documented tasks correctly and whether intended users can understand and use its scenario-based information. Technical verification and user-centered assessment will be reported separately: a passing module test does not establish usability, and a favorable perception rating does not establish geographic or institutional validity. The

ISO/IEC 25010:2023 model will be applied only through a defined instrument and analysis procedure, with the actual participant group, completed responses, scoring, and results reported after collection.

Research Limitations

The preliminary surveys are limited by their small, nonrandom or undocumented selection procedures and uncertain overlap. The second report contains design recommendations that are not implemented FloodSense capabilities. The prototype also lacks an approved operational susceptibility dataset, completed resident accounts, resident-facing nearby-center discovery, fixed-schema CSV exchange, completed formal system evaluation, and production deployment. These limitations bound the interpretation of demonstrations and future findings.

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